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MSCDDF: Multi-Stage Caption-Driven Diffusion Framework for Remote Sensing Image Semantic Segmentation

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***Abstract*—In recent years, diffusion models (a generative AI technique that creates images by progressively removing noise) show great potential in image generation. Stable Diffusion (SD) has optimized a low-dimensional latent space to balance efficiency and quality, inspiring cross-domain applications in semantic segmentation. However, when directly applied to remote sensing (RS) semantic segmentation in high-altitude scenarios, SD faces three key challenges: perspective and semantic bias, limitations of the attention mechanism (e.g., difficulty in localizing targets), and a semantic caption gap (misalignment between text and visual content). To tackle these, this paper proposes a Multi-Stage Caption-Driven Diffusion Framework (MSCDDF) with four synergistic modules. First, a remote sensing semantic-aligned caption generation (RSCG) method that bridges the text-visual gap via two-stage refinement. Second, an adaptive fine-tuning via instance cropping (AFT) strategy to reduce domain bias. Third, a multi-layer weighted attention joint generation (MWAJG) architecture to enhance segmentation accuracy. Fourth, a ground sampling distance-based synthetic data generation (GSD-SG) approach to improve dataset diversity. Experiments on iSAID and NWPU datasets demonstrate significant performance gains, including a 15.78% mIoU improvement for Mask2Former and an average 10.21% mIoU boost across multiple models, validating the framework's effectiveness in handling RS complexities. Our source code is publicly available at https://github.com/WangXin81/MSCDDF.**

***Index Terms*—remote sensing image, diffusion model, caption generation, semantic segmentation**

I. Introduction

In recent years, diffusion models have shown remarkable potential in image generation and inpainting via progressive denoising [1]. The Latent Diffusion Model, i.e., Stable Diffusion (SD) proposed by Rombach et al. [2] optimized a low-dimensional latent space to balance efficiency and quality, inspiring semantic segmentation applications. Lin et al. [3] developed SDSeg, the first latent diffusion segmentation model, enhancing single-step reverse process accuracy via latent estimation and fusion connections. Wu et al. [4] proposed DiffuMask, using SD to generate synthetic images with pixel-level masks and optimizing masks via supervised affinity networks. Wang et al. [5] leveraged SD's cross-/self-attention to generate class-specific attention maps for annotation-free mask extraction. Nguyen et al. [6] introduced Dataset Diffusion, a text-driven approach for generating synthetic images with segmentation labels to improve data quality and reduce annotation burdens.

Despite Stable Diffusion's excellence in natural image generation, its direct application to RS semantic segmentation faces three core challenges: 1) Perspective and Semantic Bias arise from SD's pre-training data being mainly human-level perspective natural images versus RS images' high-altitude overhead projections with objects minimized in complex backgrounds. 2) Attention Mechanism Limitations: In dense multi-class RS scenes with similar spectral features, SD's attention fails target localization. 3) Semantic Caption Gap: Models like BLIP2 [8] yield detailed captions for natural images (COCO [9], VOC [10]), but only basic class labels for RS datasets (iSAID [11], Vaihingen [7]).

To address the above challenges, this paper proposes a Multi-Stage Caption-Driven Diffusion Framework (MSCDDF) for RS image semantic segmentation, comprising four synergistic modules:

(1) Remote Sensing Semantic-Aligned Caption Generation (RSCG): A two-stage approach closes the RS semantic gap. First, BITA-based initial semantic mapping generates basic descriptions; then, a multi-dimensional refinement module standardizes terms, enhances multi-view details, and filters via dual validation for strong text-visual alignment.

(2) Adaptive Fine-Tuning via Instance Cropping (AFT): Cropping single-class ground objects from RS images creates low-interference sub-datasets. RSCG-generated captions detailing object attributes and spatial relations fine-tune SD, reducing semantic complexity and improving alignment for RS adaptation.

(3) Multi-Layer Weighted Attention Joint Generation (MWAJG): Its dual-stream design overcomes attention issues. The image branch keeps SD's generation ability, while the mask branch uses multi-layer weighted attention. Feature fusion from U-Net layers via cross, self, and enhanced attention generates accurate masks without degrading image quality.

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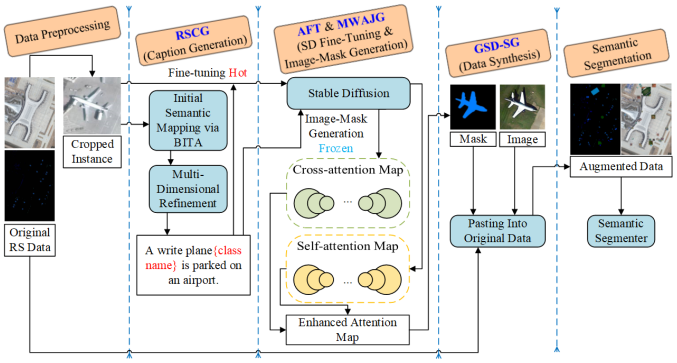


Fig. 1. Overview of the proposed MSCDDF.

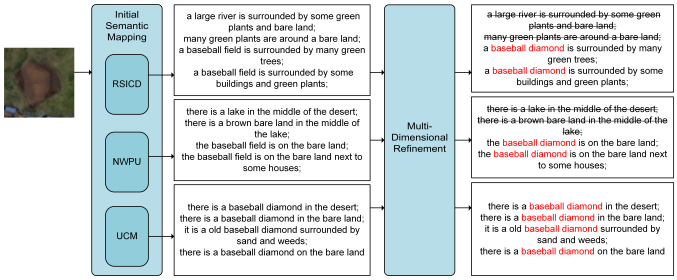


Fig. 2. Workflow of the proposed RSCG.

(4) Ground Sampling Distance-Based Synthetic Data Generation (GSD-SG): By injecting semantically constrained, cross-scale object combinations into RS images, GSD-SG boosts dataset diversity. This helps segmentation models learn multi-level features, improving generalization to complex scenes.

II. Ease of Use

Figure 1 illustrates the overall architecture of the MSCDDF framework, mainly consisting of RSCG, AFT, MWAJG, and GSD-SG.

A. Remote Sensing Semantic-Aligned Caption Generation (RSCG)

Traditional RS semantic segmentation datasets lack captions, limiting cross-modal models. While BLIP [13] and BLIP2 [8] excel on natural images, they struggle with RS data. The BITA method [12] addresses this via multi-scale frequency features and visual-text alignment, improving RS captioning over BLIP/BLIP2. However, BITA still struggles with semantic granularity and spatial context in complex scenes. To resolve this, we propose RSCG. It employs a two-stage strategy: initial semantic mapping using BITA's framework, followed by multi-dimensional refinement to enhance caption precision.

1) Initial Semantic Mapping: Leveraging BITA's interactive Fourier transformer, this stage extracts multi-scale spatial features in the frequency domain from RS images. Via image-caption contrastive learning, it generates initial captions containing basic classes and coarse spatial-frequency information. However, these captions often lack RS-specific terminology and fail to model fine-grained spatial relationships.

2) Multi-dimensional Refinement: To address BITA's limitations, we propose a multi-dimensional refinement strategy via a multi-view generation-semantic correction framework:

First, a domain-specific standard library is established for ambiguous keyword pairs (e.g., "baseball diamond" vs. "baseball field"). Post-generation, NLP-based text matching replaces non-standard terms with domain labels.

Second, a fine-grained data augmentation scheme enriches feature diversity: collecting additional samples for confusing classes (e.g., "large"/"small vehicle") and training on diverse datasets. Images are rotated (0°, 90°, 180°, 270°) during training to capture angle-dependent features, generating multi-perspective captions.

Third, a mechanism checks caption-semantic consistency to remove misclassifications, reducing errors and enhancing reliability.

3) Procedures of RSCG: Based on the two-stage strategy, RSCG follows these steps:

Step 1: Multi-view image generation. Define the angle set $\Theta = \{0°, 90°, 120°, 270°\}$. Rotate the original RS image by each $\theta \in \Theta$ to obtain a multi-view image set $\{X_\theta\}$.

Step 2: Multi-model caption generation. Let the model parameter sets be $\Phi = \{\varphi_1, \varphi_2, \varphi_3\}$, where $\varphi_1, \varphi_2, \varphi_3$ correspond to BITA parameters trained on the RSICD [14], NWPU [15], and UCM [16] datasets, respectively. Generate captions for each X_θ using all φ_i, resulting in a caption set C:

$$C = \bigcup_{\theta \in \Theta} \mathrm{B}(X_\theta; \varphi_i), \quad i = 1, 2, 3 \tag{1}$$

where $\mathrm{B}(\cdot; \varphi_i)$ denotes the BITA model with parameter φ_i.

Step 3: Semantic refinement. Filter irrelevant captions and standardize non-standard terms using a keyword library:

$$C^* = \mathrm{R}(C) \tag{2}$$

Figure 2 illustrates the workflow: 12 captions are generated via four rotations and three models, followed by filtering and standardization, ensuring robust text-visual alignment.

B. Adaptive Fine-Tuning via Instance Cropping (AFT)

To bridge domain gaps between natural image pre-trained SD models and RS images, we propose conditional fine-tuning via instance cropping and precise captioning for adaptive migration to RS scenes.

1) Diffusion Model Fundamental Framework: The core pipeline of SD consists of a bidirectional diffusion process [1]:

(1) Forward Diffusion. Encoder the RS instance image X_0 into latent features Z_0 via an encoder E:

$$Z_0 = E(X_0) \tag{3}$$

Add noise to Z_0 using a noise schedule β_t to generate noisy features:

$$Z_t = \sqrt{1 - \overline{\alpha_t}}\epsilon + \sqrt{\overline{\alpha_t}} Z_{t-1}, \overline{\alpha_t} = \prod_{i=1}^{t} (1 - \beta_i) \tag{4}$$

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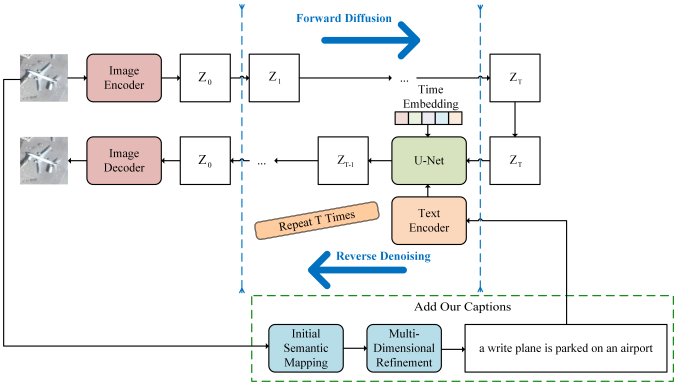


Fig. 3. Flowchart of the proposed AFT.

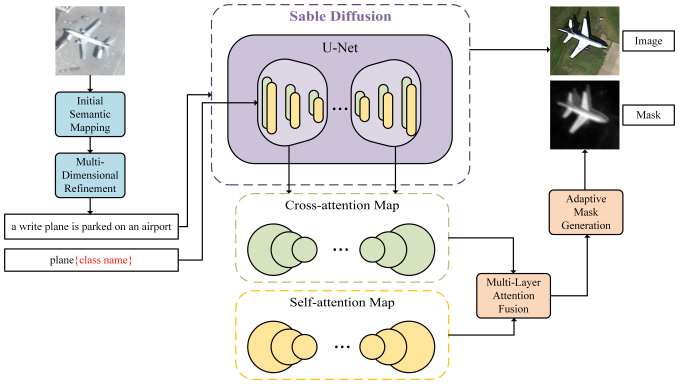


Fig. 4. Flowchart of the proposed MWAJG.

where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

(2) Reverse Diffusion. The denoising network $\epsilon_\theta(Z_t, t, y)$ approximates the inverse process to recover Z_{t-1} from Z_t, guided by caption embeddings y:

$$\mathcal{L} = \mathbb{E}_{X_0, \epsilon \sim \mathcal{N}(0,\mathbf{I}), t \sim [1,T]} \left\| \epsilon - \epsilon_\theta \left(Z_t, t, y \right) \right\|^2 \tag{5}$$

2) Remote Sensing Scene Adaptation: To address RS images' complex semantics, we propose: (1) Instance cropping - Extracts single-object sub-images to reduce complexity and capture fine-grained features. (2) Conditional fine-tuning - Uses cropped image-caption pairs: Encode images to Z_0, diffuse to Z_t, then denoise with caption embeddings to learn RS-specific distributions. 3)Multi-modal alignment - Enhances text-visual alignment via spatial prepositions and domain keywords in captions.

As shown Figure 3, the AFT is implemented as follows:

Step 1: Preprocessing. Crop single-object sub-images and generate RSCG captions with spatial descriptions.

Step 2: Forward diffusion. Encode sub-images into Z_0 and generate T-step noisy features .

Step 3: Reverse denoising. Input Z_t, time-step encoding, and caption embeddings into U-Net for restore high-fidelity RS instances.

C. Multi-Layer Weighted Attention Joint Generation (MWAJG)

Diffusion models excel in semantic/instance/panoramic segmentation via self-attention for global dependencies and cross-attention for text-visual alignment. Our joint generation framework (Figure 4) uses fine-tuned SD to generate RS images, extracts multi-layer attention features from U-Net, then generates semantic masks via weighted fusion and adaptive optimization.

1) Attention Feature Extraction: During SD's reverse denoising, U-Net's self-attention and cross-attention layers model latent feature spatial dependencies and text-semantic guidance [5].

Self-attention layers capture contextual dependencies among latent features $\mathbf{z}_t$ via similarity matrix computation.

$$\begin{aligned} \text{SelfAttn}\left(\mathbf{z}_t\right) &= \text{Softmax}\left(\mathbf{Q}\mathbf{K}^{\mathrm{T}}/\sqrt{d_k}\right), \\ \mathbf{Q} &= \mathbf{W}_q \mathbf{z}_t, \quad \mathbf{K} = \mathbf{W}_k \mathbf{z}_t \end{aligned} \tag{6}$$

where $\mathbf{W}_q$, $\mathbf{W}_k$ are projection matrices, and d_k is the query vector dimension.

Cross-attention layers establish associations between latent features and text embedding:

$$\begin{aligned} \text{CrossAttn}\left(\mathbf{z}_t, \mathbf{y}\right) &= \text{Softmax}\left(\mathbf{Q}\mathbf{K}_y^{\mathrm{T}}/\sqrt{d_k}\right)_y, \\ \mathbf{Q} &= \mathbf{W}_q \mathbf{z}_t, \quad \mathbf{K}_y = \mathbf{W}_k \mathbf{y} \end{aligned} \tag{7}$$

2) Multi-Layer Attention Fusion: To leverage complementary U-Net attention map characteristics—shallow layers for fine-grained textures, deep layers for object contours—we design a multi-layer attention fusion mechanism.

(1) Weighted Fusion of Cross-Attention Maps. We assign weights w_l to the l-th layer across-attention map $\mathbf{A}_l^{\text{cross}}$ based on semantic importance:

$$\mathbf{A}^{\text{cross}} = \sum_{l \in L} w_l \cdot \text{BilinearUp}\left(\mathbf{A}_l^{\text{cross}}, H \times W\right) \tag{8}$$

(2) Mean Fusion of Self-Attention Maps. Self-attention maps $\mathbf{A}_l^{\text{self}}$ reflect the spatial similarity of latent features. Noise is suppressed by averaging across layers:

$$\mathbf{A}^{\text{self}} = \frac{1}{|L|} \sum_{l \in L} \text{BilinearUp}\left(\mathbf{A}_l^{\text{self}}, H \times W\right) \tag{9}$$

(3) Reinforced Fusion of Attention Maps.To mitigate coarse granularity and noise in cross-attention maps, we leverage self-attention's spatial correlation modeling. By capturing pairwise dependencies between latent positions, self-attention propagates query information from initial cross-attention maps to similar locations (including non-salient regions), refining cross-attention quality.

We fuse the two maps via element-wise multiplication after raising self-attention maps $\mathbf{A}_l^{\text{self}}$ to a power k:

$$\mathbf{A}^{\text{enhanced}} = \text{Softmax}\left(\left(\mathbf{A}^{\text{self}}\right)^{\kappa} \cdot \mathbf{A}^{\text{cross}}\right) \tag{10}$$

3) Adaptive Mask Generation: To address high complexity and large intra-class variations of RS ground objects, which challenge fixed-threshold segmentation, we propose a scale-adaptive Otsu framework refining semantic masks via Gaussian noise reduction and class-specific threshold calculation.

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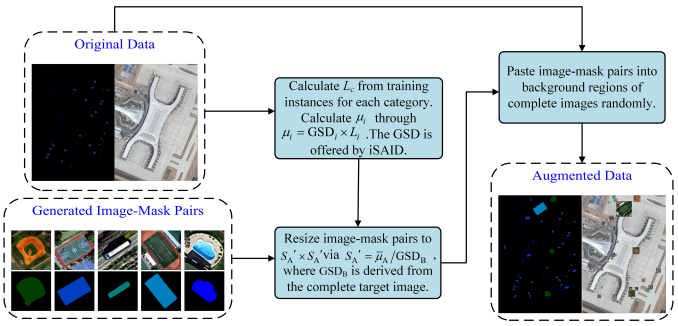


Fig. 5. Flowchart of the proposed GSD-SG.

First, to mitigate random noise in reinforced attention maps, the Gaussian filter window size ω_c is dynamically adjusted according to the target category's spatial scale: larger windows for large-scale targets, while smaller windows for small-scale targets to avoid over-blurring:

$$\overline{\mathbf{A}}c = \mathrm{G}\left(\mathbf{A}c^{\text{enhanced}}, \sigma_c\right) \tag{11}$$

where $\mathbf{A}_c^{\text{enhanced}}$ is the category-specific reinforced attention map, $\overline{\mathbf{A}}_c$ is the denoised map, and the standard deviation σ_c is automatically determined to align filtering intensity with the target's spatial scale.

Next, the Otsu method computes the optimal segmentation threshold $\overline{\mathbf{A}}_c$ from , generating a binary semantic mask M_c:

$$M_c(i,j) = \begin{cases} 1, & \text{if } \overline{\mathbf{A}}_c(i,j) \geq T_c \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where (i,j) are pixel coordinates, and the $M_c(i,j) = 1$ region denotes the target category's semantic scope. This approach offers advantages such as scale sensitivity, adaptive thresholding, and noise-detail balance.

D. Ground Sampling Distance-Based Synthetic Data Generation (GSD-SG)

Instance images train SD models to generate synthetic image-mask pairs, enhancing dataset semantic richness. For synthesis, generated pairs are composited onto random training-set backgrounds via a pasting method (Figure 5), using category-specific scaling coefficients derived from Ground Sampling Distance (GSD) [17] for adaptive scaling.

Mathematically, GSD represents the ground dimension per pixel and is defined as:

$$\mathrm{GSD} = (H \times S)/(f \times L) \tag{13}$$

where H, S, f, L represent flight height, sensor size, focal length, and image side length, respectively. In practice, H, S, f are consolidated into a single parameter μ, simplifying (13) to:

$$\mathrm{GSD} = \mu/L, \quad \mu = H \times S/f \tag{14}$$

For each category c, the average side length L_c is computed from training instances. Leveraging Eq. (14), the instance-specific coefficient $\mu_i = \mathrm{GSD}_i \times L_i$ is calculated for each instance i and averaged across all instances to obtain the category coefficient $\bar{\mu}_c$.

The average coefficient $\bar{\mu}_c$ is critical for image synthesis, ensuring generated images align with real-world scales. To add a generated pair A to a background pair B, we first query the GSD value GSD_B of B, and then retrieve the category coefficient $\bar{\mu}_A$ of A. To ensure the synthetic object's real-world dimensions align with those in the background image, the generated pair A is resized to $S'_A \times S'_A$ before being pasted into B:

$$S'_{\mathrm{A}} = \bar{\mu}_{\mathrm{A}}/\mathrm{GSD}_{\mathrm{B}} \tag{15}$$

III. Results

A. Datasets

To validate the MSCDDF framework, we utilize two RS datasets: iSAID [11] and NWPU [15].

iSAID, is a large-scale instance segmentation dataset with fine-grained annotations, containing 2,806 high-resolution RS images across 15 categories (1,411 training, 458 validation). Its pixel-level masks enable end-to-end framework evaluation.

NWPU, a 45-class scene classification benchmark with 31,500 RGB images, was extended to NWPU-Caption by Cheng et al. [15], providing text descriptions for indirect mask generation.

B. Implementation Details

Experiments use PyTorch across two platforms: 1) Image generation on local Linux (RTX 4060 Ti 16GB; i5-13600KF 32GB; CUDA 12.2, PyTorch 2.3.0). 2) Semantic segmentation on AutoDL (RTX 3090 24GB; Xeon Platinum 45GB; CUDA 12.1, PyTorch 2.3.0)

Dual setup optimizes resources: lightweight caption/mask tasks run locally for debugging; heavy segmentation uses AutoDL for parallel workflows.

Generation Settings: SD 1.5 fine-tuned @ 100k epochs (lr=1e-5).

Mmsegmentation Settings: (1)DeepLabV3Plus/FCN/Mask2Former/SegFormer [18]–[21]. (2)ResNet50/101 or MiT-B0 backbones. (3)512×512 crops, 80k epochs (Mask2Former: 90k). (4)Default hyperparameters except batch size.

C. Effects of Multimodal Data Generation

1) Caption Generation: To demonstrate RSCG's effectiveness, we generate standardized image-caption pairs for SD fine-tuning from iSAID.

From 4,241 instance images (11 classes) cropped→resized→normalized to 256×256, RSCG generates 12 semantically rich captions per image (object types/spatial contexts/scene semantics), yielding 50,892 image-caption pairs for training SD in RS-specific text-visual correlations.

Figure 6 shows coherent captions that align with RS scene semantics, validating the dataset's quality for model fine-tuning.

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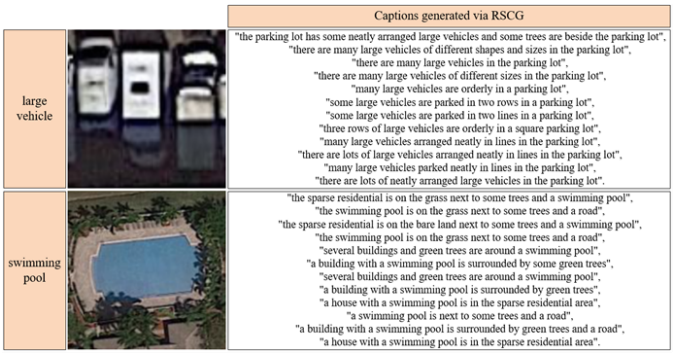


Fig. 6. Captions generated via RSCG on iSAID.

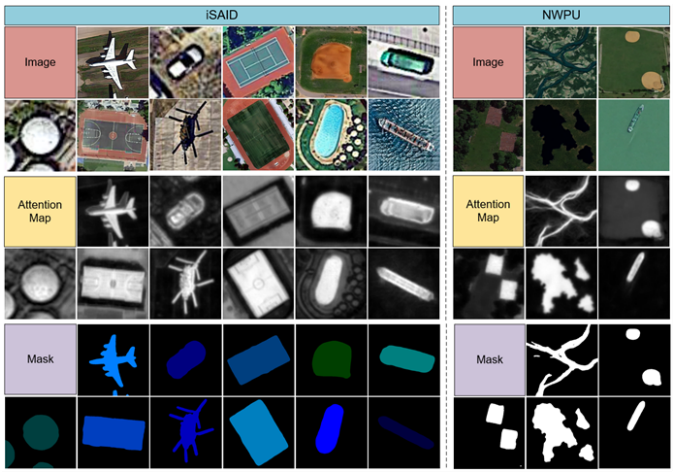


Fig. 7. Generated images with attention maps and masks on iSAID and NWPU, respectively.

2) Image-Mask Generation: Using RSCG-generated captions, we fine-tune SD to generate image-mask pairs for validating AFT and MWAJG.

For AFT, 1,632 instance images and 11,760 captions (11 classes) build a text-visual alignment space to reduce domain gaps. For MWAJG, 4,400 captions (balanced across categories, 400/class) ensure training-data independence, prevent leakage, and leverage SD's noise sampling for consistent yet varied outputs.This pipeline generates 400 image/attention map/mask pairs per class for iSAID and 200 for NWPU (Figure 7), demonstrating accurate morphology, contextual consistency, and precise masks.

3) Synthetic Data Generation: GSD-SG integrates generated instances into original images to enhance dataset diversity: upper limit and semantic label matching ensure augmented images include only absent classes for annotation consistency, while GSD resizes synthetic objects to real-world scales in backgrounds for spatial coherence. Figure 8 shows synthetic images (884×862 to 2063×1674 pixels) with natural background fusion and multi-scale adaptability, empirically validating GSD-SG's effectiveness.

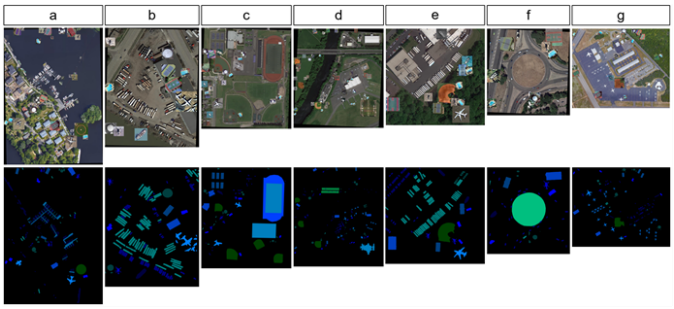


Fig. 8. Generated images with attention maps and masks on iSAID and NWPU, respectively.

D. Effects of Semantic Segmentation

1) Overall Segmentation Results: We evaluate models using Overall Accuracy (OA), Mean Intersection over Union (mIoU), and Mean Accuracy (mAcc) [22], [23]. OA measures pixel-level classification correctness. mIoU, the core semantic segmentation metric, quantifies segmentation for complex scenes, small objects, and boundaries via category-balanced overlap. mAcc assesses category balance to mitigate class bias.

Table I shows significant mIoU improvements with the augmented dataset across all architectures:

(1) DeepLabV3Plus+ResNet50/101 shows 7.62% and 14.63% gains from multi-scale feature learning in diverse scenes.

TABLE I
OVERALL SEGMENTATION RESULTS OF DIVERSE SEGMENTERS TRAINED ON THE ORIGINAL DATASET (ISAID) AND AUGMENTED DATASET (A_ISAID) WITH THE SAME TEST SET. (SE FOR SEGMENTER, BA FOR BACKBONE, TS FOR TRAINING SET, DL3+ FOR DEEPLABV3PLUS, M2F FOR MASK2FORMER, SF FOR SEGFORMER, R50 FOR RESNET50, R101 FOR RESNET101, MB0 FOR MIT-B0, I FOR ISAID, A FOR A_ISAID)

Se	Ba	Ts	OA (%)	mIoU (%)	mAcc (%)
DL3+	r50	i	97.53	35.48	45.42
		A	97.46 (-0.07)	43.10 (+7.62)	58.34 (+12.92)
	r101	i	96.42	27.74	39.21
		A	97.89 (+1.47)	42.37 (+14.63)	57.17 (+17.96)
FCN	r50	i	98.00	34.57	39.28
		A	98.39 (+0.39)	46.48 (+11.91)	52.39 (+13.11)
	r101	i	97.92	30.49	33.62
		A	98.29 (+0.37)	43.24 (+12.75)	47.52 (+13.90)
M2F	r50	i	98.18	31.75	41.27
		A	98.50 (+0.32)	47.53 (+15.78)	56.44 (+15.17)
	r101	i	98.40	38.85	47.57
		A	98.53 (+0.13)	48.02 (+9.17)	56.34 (+8.77)
SF	MB0	i	98.10	35.25	38.87
		A	98.17 (+0.07)	45.07 (+9.82)	57.33 (+18.46)

(2) FCN+ResNet50/101 shows 11.91% and 12.75% improvements from cross-resolution feature synergy.

(3) Mask2Former+ResNet50/101 shows 15.78% and 9.17% gains from high-precision masks enhancing small-object contours.

(4) SegFormer+MiT-B0 shows a 9.28% mIoU boost, balancing efficiency with fine-grained augmented images.

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TABLE II

PER-CLASS IOU (%) RESULTS OF DEEPLABV3PLUS/FCN TRAINED ON ISAID AND A_ISAID. (SH FOR SHIP, S-T FOR STORAGE-TANK, B-D FOR BASEBALL-DIAMOND, T-C FOR TENNIS-COURT, B-C FOR BASKETBALL-COURT, G-T-F FOR GROUND-TRACK-FIELD, BR FOR BRIDGE, L-V FOR LARGE-VEHICLE, S-V FOR SMALL-VEHICLE, HE FOR HELICOPTER, S-P FOR SWIMMING-POOL, R-A FOR ROUND-ABOUT, S-B-F FOR SOCCER-BALL-FIELD, PL FOR PLANE, HA FOR HABOR)

Segmenter		DeepLabV3Plus				FCN			
Backbone		ResNet50		ResNet101		ResNet50		ResNet101	
Training Set		iSAID	A_iSAID	iSAID	A_iSAID	iSAID	A_iSAID	iSAID	A_iSAID
class name	sh	45.63	54.04 (+8.41)	48.69	55.88 (+7.19)	50.36	55.59 (+5.23)	52.03	56.45 (+4.42)
	s-t	49.47	50.49 (+1.02)	37.50	54.48 (+16.98)	18.74	60.03 (+41.29)	20.28	43.96 (+23.68)
	b-d	48.87	25.92 (-22.95)	21.96	45.25 (+23.29)	41.38	52.59 (+11.21)	48.47	33.28 (-15.19)
	t-c	22.31	88.36 (+66.05)	0.52	87.71 (+87.19)	79.08	89.80 (+10.72)	67.45	89.54 (+22.09)
	b-c	25.22	44.27 (+19.05)	11.20	31.89 (+20.69)	23.55	20.17 (-3.38)	4.80	50.51 (+45.71)
	g-t-f	17.39	21.74 (+4.35)	27.85	34.26 (+6.41)	3.63	17.76 (+14.13)	3.26	8.98 (+5.72)
	br	4.72	12.29 (+7.57)	10.74	11.65 (+0.91)	19.21	16.86 (-2.35)	16.00	17.36 (+1.36)
	l-v	40.50	53.80 (+13.30)	34.27	51.61 (+17.34)	47.52	51.97 (+4.45)	47.13	43.70 (-3.43)
	s-v	39.55	38.02 (-1.53)	28.52	36.84 (+8.32)	35.64	40.52 (+4.88)	25.04	36.02 (+10.98)
	he	0.00	4.09 (+4.09)	0.00	3.21 (+3.21)	0.00	20.20 (+20.20)	0.00	4.09 (+4.09)
	s-p	49.97	39.31 (-10.66)	27.60	9.09 (-18.51)	37.04	46.64 (+9.60)	19.88	40.01 (+20.13)
	r-a	26.76	29.62 (+2.86)	20.81	6.12 (-14.69)	8.92	27.53 (+18.61)	1.68	32.06 (+30.38)
	s-b-f	37.94	28.87 (-9.07)	15.94	40.29 (+24.35)	4.55	47.71 (+43.16)	2.93	29.63 (+26.70)
	pl	39.85	65.89 (+26.04)	41.21	63.25 (+22.04)	63.62	65.31 (+1.69)	54.70	68.37 (+13.67)
	ha	21.74	35.28 (+13.54)	20.17	48.23 (+28.06)	21.77	32.48 (+10.71)	26.16	39.45 (+13.29)

TABLE III

PER-CLASS IOU (%) RESULTS OF MASK2FORMER/SEGFORMER TRAINED ON ISAID AND A_ISAID.

Segmenter		Mask2Former				SegFormer	
Backbone		ResNet50		ResNet101		MiT-B0	
Training Set		iSAID	A_iSAID	iSAID	A_iSAID	iSAID	A_iSAID
class name	ship	59.96	61.94 (+1.98)	61.78	65.66 (+3.88)	49.63	44.41 (-5.22)
	storage-tank	48.72	66.04 (+17.32)	60.08	66.11 (+6.03)	35.74	54.56 (+18.82)
	baseball-diamond	0.00	26.91 (+26.91)	0.00	25.10 (+25.10)	17.99	41.40 (+23.41)
	tennis-court	44.25	87.61 (+43.36)	69.66	88.83 (+19.17)	77.47	76.82 (-0.65)
	basketball-court	0.00	26.82 (+26.82)	0.00	24.27 (+24.27)	37.93	40.07 (+2.14)
	ground-track-field	4.80	10.47 (+5.67)	13.57	10.60 (-2.97)	9.99	32.88 (+22.89)
	bridge	9.90	12.08 (+2.18)	12.53	11.39 (-1.14)	0.61	4.04 (+3.43)
	large-vehicle	53.87	54.39 (+0.52)	56.00	58.49 (+2.49)	47.76	49.90 (+2.14)
	small-vehicle	44.54	46.28 (+1.74)	50.54	50.22 (-0.32)	32.72	43.70 (+10.98)
	helicopter	0.00	17.09 (+17.09)	0.00	18.15 (+18.15)	0.00	21.79 (+21.79)
	swimming-pool	9.72	60.46 (+50.74)	30.43	59.55 (+29.12)	29.15	47.71 (+18.56)
	round-about	0.03	11.01 (+10.98)	5.47	10.93 (+5.46)	4.11	4.03 (-0.08)
	soccer-ball-field	10.43	52.19 (+41.76)	33.78	51.78 (+18.00)	30.19	62.45 (+32.26)
	plane	73.26	77.14 (+3.88)	77.43	77.77 (+0.34)	59.99	63.90 (+3.91)
	harbor	49.94	51.42 (+1.48)	51.72	50.69 (-1.03)	32.57	35.13 (+2.56)

TABLE IV

ABLATION STUDIES ON RSCG.

Segmenter	**Backbone**	**BITA**	**Refinement**	**Test**		
				OA (%)	mIoU (%)	mAcc (%)
SegFormer	MiT-B0			97.39	38.91	48.62
		✓		97.86	40.04	49.44
		✓	✓	**98.17**	**45.07**	**57.33**

TABLE V

IMAGE QUALITY IMPACT ON SEMANTIC SEGMENTATION. (↑ INDICATES HIGHER VALUES ARE PREFERABLE, ↓ INDICATES LOWER VALUES ARE PREFERABLE. OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD.)

BITA	**Refinement**	**IS (↑)**	**FID(↓)**	**Segmenter**	**Backbone**	**Test**		
						OA (%)	**mIoU (%)**	**mAcc (%)**
		11.5115	96.76371834	SegFormer	MiT-B0	97.39	38.91	48.62
✓		11.6660	99.12600068			97.86	40.04	49.44
✓	✓	11.9993	77.72580724			**98.17**	**45.07**	**57.33**

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2) Class-Specific Segmentation Results: Table II-Table III detail IoU performance across 15 iSAID categories.

First, over 85% of categories, including complex classes (bridge, harbor) and small targets (small vehicle, helicopter), show improvements. Notable gains include DeepLabV3Plus+ResNet101 improving tennis-court IoU from 0.52% to 87.71%. Consistency across CNN and Transformer architectures demonstrates the strategy's architecture-agnostic efficacy from synthetic data's diverse scene contexts.

Second, four categories without direct augmentation (ground-track-field, bridge, round-about, harbor) achieve 0.91%–28.06% IoU improvements. For example, SegFormer+MiT-B0 increases ground-track-field IoU from 9.99% to 32.88%, enabled by cross-category semantic correlations in RSCG captions embedded in synthetic images. This highlights scene-level semantic learning beyond traditional augmentation's sample-quantity reliance.

Third, approximately 15% of categories show IoU degradation (e.g., swimming-pool, ship), primarily from synthetic samples failing to capture unique feature distributions; these marginal cases highlight opportunities for category-specific augmentation refinement.

IV. Discussion

A. Ablation Study

The RSCG framework uses BITA for initial captioning, followed by multi-dimensional refinement for enhanced RS-specific semantic fidelity. To isolate contributions, two ablation experiments control single variables:

1) replacing BITA with fixed-template captions ("bird view of class name") while keeping other processes unchanged;

2) retaining BITA's raw captions without refinement. Table IV quantifies incremental improvements: for mIoU, BITA alone improves 1.13% over fixed templates, with refinement adding another 5.03%; for OA, BITA contributes 0.74%, refinement adds 0.31%; for mAcc, BITA improves 0.82%, refinement adds 7.89%.

These results confirm BITA's semantic grounding and refinement are indispensable for maximizing segmentation accuracy, especially in category-balanced metrics.

B. Convergence Speed Analysis

Figure 9 illustrates the augmented dataset's impact on convergence speed across four models.

Key observations: (1) All models except Mask2Former+ResNet101 accelerate validation mIoU growth with augmented data. (2) SegFormer+MiT-B0 (Figure 9(a)) maintains consistent mIoU superiority per iteration, highlighting the augmented dataset's capability to expedite feature learning. (3) Mask2Former's initial underperformance (15k iterations) may stem from its reliance on dense mask supervision—initially difficult to replicate with synthetic data—yet eventually converges to higher mIoU than the original dataset.

These trends validate that multimodal augmentation not only improves peak performance but also accelerates training efficiency, aligning with diffusion models' capacity to inject diverse semantic contexts.

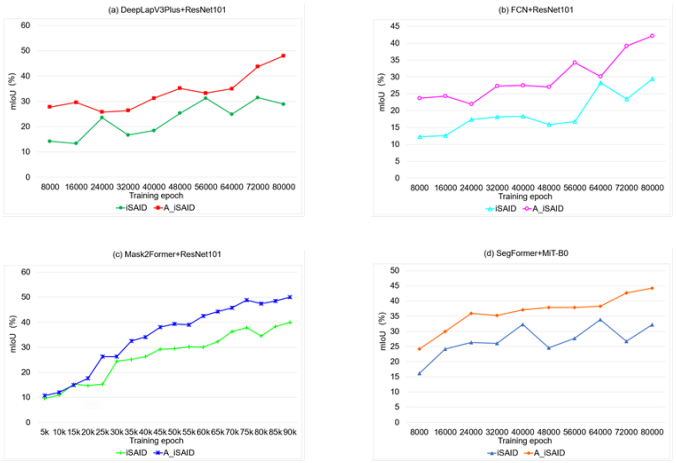


Fig. 9. Convergence speed of diverse segmenters trained on iSAID and A_iSAID. (a) DeepLabV3Plus+ResNet101; (b) FCN+ResNet101; (c) Mask2Former+ResNet101; (d) SegFormer+MiT-B0.

C. Image Quality Impact on Semantic Segmentation

We access synthetic image quality via Inception Score (IS) and Fréchet Inception Distance (FID), correlating metrics with segmentation performance. IS measures semantic consistency and diversity via category entropy, while FID quantifies feature distribution similarity to real data using InceptionV3 embeddings [24].

Table V shows task-specific tradeoffs: (1) Model 1 to Model 2 prioritizes diversity over authenticity. BITA elevates IS from 11.51 to 11.67, enriching intra-class variation. Despite FID increasing to 99.13 due to long-tail background features, mIoU/OA improve significantly—confirming task-relevant semantics outweigh global fidelity. (2) Model 2 to Model 3 achieves synergistic optimization: Refinement further boosts IS to 11.99, reduces FID to 77.73, and balances diversity/authenticity. This dual improvement drives peak mIoU/OA, proving high-quality synthetic data requires both semantic richness and distributional alignment.

These findings underscore that while semantic diversity is critical for generalization, alignment with real-world feature distributions remains essential for optimal segmentation accuracy.

V. Conclusions

This study addresses Stable Diffusion adaptation challenges for high-altitude remote sensing semantic segmentation, where perspective bias, attention limitations, and text-visual gaps limit performance. We introduce the Multi-Stage Caption-Driven Diffusion Framework (MSCDDF), integrating domain-specific alignment, adaptive fine-tuning, and augmentation strategies. Its four modules—RSCG, AFT, MWAJG, and GSD-SG—collectively mitigate domain limitations, empowering SD to model RS complexities. Evaluations across RS datasets

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validate MSCDDF's efficacy for segmentation with complementary data generation benefits. Future work may integrate multi-modal RS data and optimize for edge-device real-time inference, enhancing diffusion models in geospatial analytics.

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